Amama Mahmood

Department of Computer Science
Johns Hopkins University
3400 North Charles Street,
Baltimore, MD 21218, USA

amama.mahmood@jhu.edu
https://amamamahmood.com/
Google Scholar

Research Interests

My research lies at the intersection of HCI, AI, and health. I design and evaluate LLM-powered conversational agents that support health and well-being, from personal health at home to clinicians' decision-making. Using human-centered and participatory design, I co-design these systems with patients, clinicians, and other stakeholders, and study them through long-term, in-the-wild deployments. I also examine how AI is integrated across health systems to inform responsible adoption.

Keywords: Human-Computer Interaction, Human-AI Interaction, Human-Robot Interaction, Conversational Agents, Large Language Models, Human-Subjects Research, Health & Well-Being

Current Position

Postdoctoral Fellow — September 2025–present
Malone Center for Engineering in Healthcare, Johns Hopkins University — Baltimore, MD, USA
Advisors: Chien-Ming Huang, Ziang Xiao, and Paul Yi

Education

Ph.D. in Computer Science — 2020–2025
Johns Hopkins University — Baltimore, MD, USA
Advisor: Chien-Ming Huang
Thesis: *Designing Conversation Experience: From Traditional to LLM-Powered Voice Assistants*

M.S.E. in Computer Science — 2020–2022
Johns Hopkins University — Baltimore, MD, USA

M.S.E. in Robotics — 2018–2020
Johns Hopkins University — Baltimore, MD, USA

B.E. in Electrical Engineering — 2013–2017
National University of Sciences & Technology (NUST) — Islamabad, Pakistan

Honors and Awards

Malone Postdoctoral Fellowship, Johns Hopkins University — 2025–2026
Creel Family Engineering Fellowship, Johns Hopkins University — 2020–2021
Computer Science Department Fellowship, Johns Hopkins University — 2020–2021
Fulbright Scholar — 2018–2020

Publications

* denotes equal contribution.

Manuscripts Under Review and In Preparation

M.5 **Artificial Intelligence Benefits Evaluation Framework (*AI BEF 1.0*)**
Amama Mahmood, Shiye Cao, Laura Fleig, Theodore Jensen, Yee-Yin Choong, Kristen Greene, and Chien-Ming Huang
In preparation (2026)

M.4 **Ninety-Six Ways to Measure a Scribe: Fragmented Evaluation of Ambient AI Scribes in Healthcare**

Amama Mahmood, Shiye Cao, Laura Fleig, Junan Jiang, Theodore Jensen, Yee-Yin Choong, Kristen Greene, and Chien-Ming Huang
Under review (2026)

M.3 **Guided Reflection for Personal Sleep Insight in Everyday Sleep Tracking**
Bokyung Kim, **Amama Mahmood**, Honghao Zhao, Molly Atwood, Luis Buenaver, Ziang Xiao, and Chien-Ming Huang
Under review (2026) | arXiv:2610.10822

M.2 **Better Adherence, Richer Context: A Field Evaluation of LLM-Powered Conversational Voice Diaries for Sleep**
Amama Mahmood*, Bokyung Kim*, Honghao Zhao, Molly Atwood, Luis Buenaver, Michael Smith, and Chien-Ming Huang
Under review (2026) | arXiv:2606.18596

M.1 **Conversational Agents in Behavioral Sleep Medicine: Designing Self-Report and Analytics Tools**
Amama Mahmood, Bokyung Kim, Honghao Zhao, Molly Atwood, Luis Buenaver, Michael Smith, and Chien-Ming Huang
Under review (2026) | arXiv:2509.15378

Peer-Reviewed Journal Articles

J.7 **Situated Understanding of Errors in Older Adults' Interactions with Voice Assistants: A Month-Long, In-Home Study**
Amama Mahmood, Junxiang Wang, and Chien-Ming Huang
ACM Transactions on Accessible Computing (TACCESS) (2026) | Volume 19, Issue 1 | https://doi.org/10.1145/3796236
Invited to present at ASSETS'26

J.6 **"Mango Mango, How to Let The Lettuce Dry Without A Spinner?": Exploring User Perceptions of Using An LLM-Based Conversational Assistant Toward Cooking Partner**
Szeyi Chan, Jiachen Li, Bingsheng Yao, **Amama Mahmood**, Chien-Ming Huang, Holly Jimison, Elizabeth D. Mynatt, and Dakuo Wang
Proceedings of the ACM on Human-Computer Interaction (CSCW) (2025) | Volume 9, Issue 7 | https://doi.org/10.1145/3757442

J.5 **User Interaction Patterns and Breakdowns in Conversing with LLM-Powered Voice Assistants**
Amama Mahmood, Junxiang Wang, Bingsheng Yao, Dakuo Wang, and Chien-Ming Huang
International Journal of Human-Computer Studies (IJHCS) (2025) | Volume 195 | https://doi.org/10.1016/j.ijhcs.2024.103406

J.4 **Care to Explain? AI Explanation Types Differentially Impact Physician Diagnostic Performance and Trust in AI**
Drew Prinster*, **Amama Mahmood***, Suchi Saria, Jean Jeudy, Cheng Ting Lin, Paul Yi, and Chien-Ming Huang
Radiology (2024) | Volume 313, Issue 2 | https://doi.org/10.1148/radiol.233261

J.3 **Gender Biases in Error Mitigation by Voice Assistants**
Amama Mahmood and Chien-Ming Huang
Proceedings of the ACM on Human-Computer Interaction (CSCW) (2024) | Volume 8, Issue CSCW1, Article 60, Pages 1–27 | https://doi.org/10.1145/3637337

J.2 **Crowdsourcing Thumbnail Captions: Data Collection and Validation**
Carlos Aguirre*, Shiye Cao*, **Amama Mahmood**, and Chien-Ming Huang
ACM Transactions on Interactive Intelligent Systems (TIIS) (2023) | Volume 13, Issue 3, Article 14, Pages 1–28 | https://doi.org/10.1145/3589346
Invited article following IUI'22 (C.4)

J.1 **SmartSIM—A Virtual Reality Simulator for Laparoscopy Training Using a Generic Physics Engine**
Zohaib Amjad Khan, Nabeel Kamal, Asad Hameed, **Amama Mahmood**, Rida Zainab, Bushra Sadia, Shamyl Bin Mansoor, and Osman Hasan
The International Journal of Medical Robotics and Computer Assisted Surgery (2017) | Volume 13, Issue 3, Pages e1771 | https://doi.org/10.1002/rcs.1771

Peer-Reviewed Conference Papers

C.9 **ERR@HRI 3.0 Challenge: Multimodal Detection of Errors and Anticipation in Human-Robot Interactions**
Maria T. Parreira, Micol Spitale, Maia Stiber, Shiye Cao, **Amama Mahmood**, Chien-Ming Huang, Hatice Gunes, and Wendy Ju
Proceedings of the 28th International Conference On Multimodal Interaction (ICMI) (2026) (Short paper) | Pages 1381-1385 | https://doi.org/10.1145/3776574.3833403

C.8 **ERR@HRI 2.0 Challenge: Multimodal Detection of Errors and Failures in Human-Robot Conversations**
Shiye Cao, Maia Stiber, **Amama Mahmood**, Maria T. Parreira, Wendy Ju, Micol Spitale, Hatice Gunes, and Chien-Ming Huang
Proceedings of the 33rd ACM International Conference on Multimedia (ACM-MM) (2025) (Short paper) | Pages 14130–14135 | https://doi.org/10.1145/3746027.3762073

C.7 **Interruption Handling for Conversational Robots**
Shiye Cao, Jiwon Moon, **Amama Mahmood**, Victor Antony, Ziang Xiao, Anqi Liu, and Chien-Ming Huang
Proceedings of the 2025 Robotics: Science and Systems Conference (RSS) (2025) | https://doi.org/10.48550/arXiv.2501.01568 | Acceptance rate: 27.4%

C.6 **Voice Assistants for Health Self-Management: Designing for and with Older Adults**
Amama Mahmood, Shiye Cao, Maia Stiber, Victor Antony, and Chien-Ming Huang
Proceedings of the 2025 ACM Conference on Human Factors in Computing Systems (CHI) (2025) | https://doi.org/10.1145/3706598.3713839 | Acceptance rate: 25.1%

C.5 **Effects of Rhetorical Strategies and Skin Tones on Agent Persuasiveness in Assisted Decision-Making**
Amama Mahmood and Chien-Ming Huang
Proceedings of the 2022 ACM International Conference on Intelligent Virtual Agents (IVA) (2022) | Article 7, Pages 1–8 | https://doi.org/10.1145/3514197.3549628

C.4 **Crowdsourcing Thumbnail Captions Using Time-Constrained Methods**
Carlos A. Aguirre, **Amama Mahmood**, and Chien-Ming Huang
Proceedings of the 2022 ACM International Conference on Intelligent User Interfaces (IUI) (2022) | Pages 36–48 | https://doi.org/10.1145/3490099.3511136 | Acceptance rate: 24.5%

C.3 **Owning Mistakes Sincerely: Strategies for Mitigating AI Errors**
Amama Mahmood, Jeanie W. Fung, Isabel Won, and Chien-Ming Huang
Proceedings of the 2022 ACM Conference on Human Factors in Computing Systems (CHI) (2022) | Article 578, Pages 1–11 | https://doi.org/10.1145/3491102.3517565 | Acceptance rate: 24.7%

C.2 **Visual Monitoring and Servoing of a Cutting Blade during Telerobotic Satellite Servicing**
Amama Mahmood, Balazs P. Vagvolgyi, Will Pryor, Louis L. Whitcomb, Peter Kazanzides, and Simon Leonard
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (2020) | Pages 1903–1908 | https://doi.org/10.1109/IROS45743.2020.9341485

C.1 **Classification of Multi-class Motor Imagery EEG Using Four Band Common Spatial Pattern**
Amama Mahmood, Rida Zainab, Rushda Basir Ahmad, Maryam Saeed, and Awais Mehmood Kamboh
39th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC) (2017) | Pages 1034–1037 | https://doi.org/10.1109/EMBC.2017.8037003

Refereed Symposium Papers, Workshop Papers, and Conference Abstracts

S.1 **From Our Lab to Their Homes: Learnings from Longitudinal Field Research with Older Adults**
Amama Mahmood and Chien-Ming Huang
AAAI Fall Symposium on Aging in Place (2024) | arXiv:2409.15495

A.1 **Care To Explain? Differential Impacts of Explanation Types On Physician Trust In AI**
Drew Prinster*, **Amama Mahmood***, Suchi Saria, Jean Jeudy, Cheng Ting Lin, Paul Yi, and Chien-Ming Huang
2023 Conference on Machine Intelligence in Medical Imaging (Society for Imaging Informatics in Medicine) | Podium presentation (conference abstract)

W.1 **How Mock Model Training Enhances User Perceptions of AI Systems**
Amama Mahmood, Gopika Ajaykumar, and Chien-Ming Huang
Human-Centered AI (HCAI) Workshop at NeurIPS (2021) | arXiv:2111.08830

Theses

T.2 **Designing Conversation Experience: From Traditional to LLM-Powered Voice Assistants**
Amama Mahmood
Department of Computer Science, Johns Hopkins University (2025)
Doctor of Philosophy (Ph.D.) | Thesis

T.1 **Robot Assisted 3D Block Building to Augment Spatial Visualization Skills in Children—An Exploratory Study**
Amama Mahmood
Laboratory for Computational Sensing and Robotics, Johns Hopkins University (2020)
Master of Science in Engineering (M.S.E.) Robotics | Thesis

Invited Talks

Designing Conversational Agents to Empower People and Promote Well-Being
Department of Computer Science, George Mason University — May 2025
Human-Computer Interaction (HCI) Brown Bag & CS Seminar, Johns Hopkins University — April 2025
Human-Centered AI Class, Northeastern University — February 2025

Human-AI Interaction for Healthcare: Through the Lens of Radiology
Annual Meeting of the Society for Imaging Informatics in Medicine (SIIM) — May 2025
Session: *Bridging the Gap: Mitigating Bias, Building Trust, and Mastering Human-AI Collaboration*

Teaching

Course Instructor, EN.601.491/691 Human-Robot Interaction — Spring 2025
Department of Computer Science, Johns Hopkins University
Overall quality: 4.11/5.00 (responses = 29, size = 30)
Teaching effectiveness: 4.21/5.00 (responses = 29, size = 30)

Teaching Assistant, EN.601.490/690 Introduction to Human-Computer Interaction — Fall 2021
Department of Computer Science, Johns Hopkins University

Course Assistant, EN.601.491/691 Human-Robot Interaction — Spring 2020
Department of Computer Science, Johns Hopkins University

Mentoring

Student Mentor — 2020–present
Johns Hopkins University
Mentored undergraduate and graduate students; co-authored papers: M.1, M.2, M.3, M.4, J.5, J.7, C.3, C.7

Human-Computer Interaction Reading Group Organizer — 2020–2023
Intuitive Computing Lab, Johns Hopkins University

Service

Peer Review

2020–present

ACM Conference on Human Factors in Computing Systems (CHI) (multiple special mentions)
ACM Transactions on Human-Robot Interaction (THRI)
International Journal of Human-Computer Studies (IJHCS)
ACM/IEEE International Conference on Human-Robot Interaction (HRI)
ACM International Conference on Multimodal Interaction (ICMI)

Outreach and Community

Organizer, Community Outreach 2024, 2025

Organized a community outreach exhibition at a local senior living center
Organized workshops for children with special needs

Organizer, Lab Hackathon Summer 2023

Organized hackathon for research group on integrating LLMs into voice assistants and robots

Member, Robotics Graduate Student Association 2020–2024

Johns Hopkins University

References

Dr. Chien-Ming Huang (Ph.D. Advisor)	John C. Malone Associate Professor, Department of Computer Science, Johns Hopkins University	chienming.huang@jhu.edu
Dr. Dakuo Wang	Associate Professor, Khoury College of Computer Sciences and the College of Arts, Media and Design, Northeastern University	d.wang@northeastern.edu
Dr. Paul Yi	Associate Member, Department of Radiology, St. Jude Children's Research Hospital	paul.yi@stjude.org
Dr. Molly Atwood	Director of Clinical Training, Behavioral Sleep Medicine Program, Assistant Professor of Psychiatry and Behavioral Sciences, School of Medicine, Johns Hopkins University	matwood4@jhmi.edu
Dr. Ziang Xiao	Assistant Professor, Department of Computer Science, Johns Hopkins University	ziang.xiao@jhu.edu